

Formal Methods for Comprehensive Testing of BGP Link-State (BGP-LS)

Problem Statement

BGP Link State (BGP-LS) is a northbound information exchange protocol extension. It allows external clients to subscribe to and collect link-state data (topology, traffic engineering metrics, segment routing SIDs, etc.). Originally designed as an export mechanism, BGP-LS is being increasingly used as a critical feed-back component in the operation and management of cloud network infrastructures. For example, TE controllers may rely on such information to respond to real-time connectivity and traffic status. Thus, the accuracy and reliability of BGP-LS is of foundational importance.

Mentorship Objectives

The objective of this mentorship is to develop systematic tests and validations for the BGP-LS implementation. More specifically, it needs to address complexities at various levels to achieve high coverage of the implementation. For example, at the high level, the stateful nature of the protocol extension itself, as well as the dynamic, disaggregated changes in a router's internal states, require thorough exploration of the handling of diverse interaction scenarios and even inconsistent states. At the low level, data processing utilities, such as the parsing of internal routing data formats, also requires systematic testing to validate the handling of both normal and abnormal data.

Expected Outcomes and Deliverables

- A set of formal models describing both the expected high-level interactions and low-level processing logics of BGP-LS.
- Benchmarks containing diverse interaction sequences and input data, providing comprehensive behavior and code coverage.
- Testing infrastructures to execute, evaluate, and refine these tests.

Recommended Qualifications

- Proficiency in Python and C/C++
- Basic knowledge of IP networking and routing
- Basic understanding of distributed systems and concurrency
- Ability to navigate and understand complex codebases

Mentor(s) Names and Contact Info

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Time and location

- Summer - Fall (6 months)
- Location: China

Repo: <https://github.com/mengqiliu20/sonic-formal>